RMIT UNIVERSITY

CSIRO

RMIT Quantum Summer School 2026

Project

Grover's Algorithm

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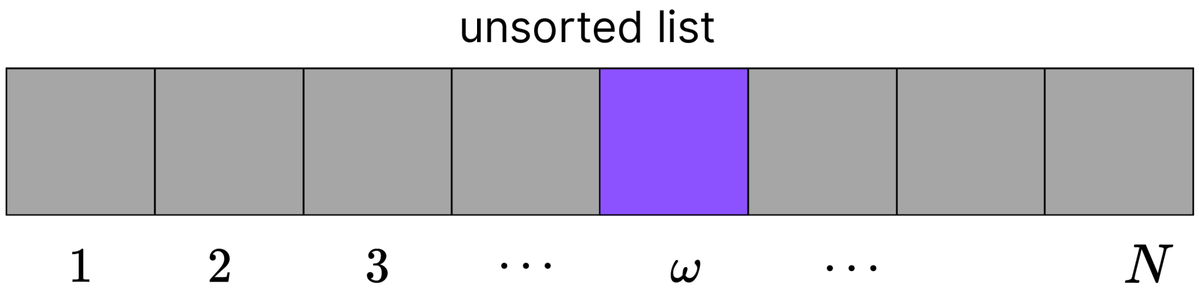
Quantum Systems, CSIRO's Data61

The Unstructured Search Problem



Given an unsorted list with N = 2^n items

Exactly one *(or a few)* **is the "marked item"** ω

How can we find ω efficiently?

→ **an unstructured search problem:**

No ordering, no structure to exploit

(finding a needle in a haystack)

→ **Classical algorithm need on average** $N/2 = O(N)$ **times (the worst case: N)**

Cannot do better than linear search in the black-box (oracle) model

In quantum computing, we can use Grover's algorithm → need $O(\sqrt{N})$**times** → **Quadratic speed up**

Project: Grover’s Algorithm

Project Overview

In this project, you will implement and analyse **Grover's quantum search algorithm** for unstructured search in **n-qubit** systems. You will start with the classical case of searching for a **single marked item**, then extend your implementation to handle **k > 1 marked items**

Deliverables

1. Multiple **oracle** functions for different marked items,
2. Working implementation of the n-qubit single-item Grover’s search
3. An extended circuit implementation supporting **k = 2, 3, 4 marked items**
4. Analysis of the **optimal iteration count** vs. **k** relationship
5. Analysis showing how the Grover operator **amplifies target amplitudes**
6. Testing and validation on ideal and noisy quantum simulators